They Like to Move It, Move It — Determining the Viscosity of Colloidal Suspensions of Monodisperse Polystyrene Microspheres in Water Using Brownian Motion

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Brownian motion is the phenomenon of microscopic particles suspended in fluid moving randomly. We use the observation of particles undergoing Brownian motion to determine the viscosity of a colloidal suspension. We use a suspension of monodisperse polystyrene microspheres in water to track movement of the particles and through that determine the coefficient of diffusion of our sample. For the particloe tracking we use the python package trackpy. Using the Stokes-Einstein relation given by $D = \frac{k_B T}{6\pi\eta r}$, where D is the diffusion coeffcient in m^2/s, $k_B = 1.38 \times 10^{-23} J/K$ is the Boltzmann constant, T is the temperatur ein Kelvin, η is the dynamic viscosity in $mPa \cdot s$ and r is the radius of the particle, we then relate the diffusion coefficient to the viscosity. We measured the Diffusion coefficient of 25 videos across 4 different particle radii $r \in \{0.373, 0.53, 0.77, 1.03\}\mu m$. We found a mean value of viscosity for the different aprticle radii as $\eta_{r=0.373} = 2.7 \pm 0.1 mPa \cdot s, \eta_{r=0.53} = 2.1 \pm 0.2 mPa \cdot s, \eta_{r=0.77} = 2.2 \pm 0.2 mPa \cdot s, \eta_{r=1.03} = 2.0 \pm 0.4 mPa \cdot s$. Perfroming a constant function fit through the data yielded a viscosity of $\eta = 2.4 \pm 0.1 mPa \cdot s$. The difficulties of the experiment and subsequently errors in measurements were influenced by inability to dierctly measure sample temperature as only room temperature measurements were performed, incosnsistent sample lighting, sample interference that created phase contrast transitions and forced us to de-focus the samples to prevent change from dark to light colour, and finally the tracking of the particles.

I. INTRODUCTION

Brownian motion describes the continuous, random movement of microscopic particles suspended in a fluid. The phenomenon was first observed by Robert Brown in the nineteenth century, who noticed the irregular movement of pollen grains suspended in water. Its theoretical foundations were subsequently developed by Albert Einstein providing a physical explanation for the observed motion and establishing its connection to the microscopic properties of fluids [1].

Einstein's theoretical description relates the mean squared displacement of suspended particles to the diffusion coefficient and elapsed time [2]. This relationship provides a way to investigate the properties of a fluid through the motion of particles suspended within it. In particular, the Stokes-Einstein relation relates the diffusion coefficient to the particle size, fluid viscosity and temperature [3]. Consequently, by tracking the Brownian motion of particles with known properties, the viscosity of the surrounding fluid can be determined.

In this experiment, Brownian motion is used to determine the viscosity of aqueous suspensions containing monodisperse polystyrene microspheres of different sizes. By tracking the displacement of individual particles over time, their diffusion coefficients can be determined and related to the viscosity of the surrounding fluid. The main objective is to investigate whether this method provides a reliable determination of viscosity and to examine how the measured viscosity depends on the size of the suspended microspheres. If successfull, this would allow an *in situ*[4] measurement of the viscosity of the liquid. This is especially advantageous since it allows for measurements without disturbing the microsystem [5].

II. THEORY

A key relation connecting Brownian motion to the fluid's viscosity is the Stokes–Einstein relation, which is given as follows

$$D = \frac{k_B T}{6\pi\eta r}, \tag{1}$$

where D in m^2/s is the diffusion coefficient, k_B is the Boltzmann constant given as $1.38 \times 10^{-23}\,\mathrm{J/K}$, T is the thermodynamic temperature in Kelvin, η is the dynamic viscosity in Pascal-seconds, and r is the particle radius in meters [2]. It applies to spherical particles suspended in a Newtonian fluid and provides us with a way to determine the viscosity of the fluid by measuring the diffusion coefficient of the suspended particles. These conditions are mostly fufilled by our setup. We can rewrite this equation in terms of the viscosity as

$$\eta = \frac{k_B T}{6\pi D r}. \tag{2}$$

To get the diffusion coefficient as a function that we can measure, we look into the movement of particles that undergo Brownian motion. To begin, we assume that the motion of each distinct particle is independent of the motion of the other suspended particles. If we then also impose the condition that the number of particles per unit volume ν is a function of only x and t, i.e. $\nu =$

$f(x,t)$, we get that it sattisfies the equation

$$\frac{\partial f}{\partial t} = D\frac{\partial^2 f}{\partial x^2}, \tag{3}$$

where D is the diffusion coefficient [2].
We then define $f(x,t)dx$ to denote the number of particles that have moved by a displacement between x and $x+dx$ in the x direction in the time interval $[0,t]$. We can impose the initial and boundary conditions as

$$f(x,0) = 0,$$
$$\int_{-\infty}^{\infty} f(x,t)dx = n,$$

where n is the total number of particles suspended in the liquid [2]. The solution to $f(x,t)$ can in this case be found analytically and is given by

$$f(x,t) = \frac{n}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}. \tag{4}$$

We can then determine the average displacement of a particle along the x direction as

$$\lambda_x = \sqrt{\langle x^2 \rangle} = \sqrt{2Dt}, \tag{5}$$

where $\langle x^2 \rangle$ is the arithmetic mean of squares of displacements along the x direction [2].
In the 2D case, we take the effect to be independent in the x and y axes. This then gives us an expression for the mean squared displacement in the xy-plane as

$$\langle R^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle = \langle x^2 + y^2 \rangle = 2Dt + 2Dt = 4Dt. \tag{6}$$

We can then use this relation to find an equation for the diffusion coefficient as

$$4Dt = \langle R^2 \rangle, \tag{7}$$
$$\frac{d}{dt}4Dt = \frac{d}{dt}\langle R^2 \rangle, \tag{8}$$
$$4D = \frac{d}{dt}\langle R^2 \rangle, \tag{9}$$
$$D = \frac{d\langle R^2 \rangle}{4dt}. \tag{10}$$

We can then plug this expression into Eqn. 2 to get

$$\eta = \frac{k_B T}{6\pi D a} = \frac{2k_B T}{3\pi \frac{d\langle R^2 \rangle}{dt} a}. \tag{11}$$

Calculating the uncertainty of a quantity is done using the formula

$$\sigma_A = A \cdot \sqrt{\sum_n \sigma_n^2}, \tag{12}$$

where A represents the quantity we want to calculate the resulting ucnertainty for and σ_n are relative uncertainties of quantities that are in the formula for A.

III. METHODOLOGY

As samples to be analysed, we are using monodisperse polystyrene microspheres of the sizes of $0.5 - 2\mu$m who are suspensed in water [6]. The resulting liquid is a collodial suspension, which is the description for mixtures where the collodial particles are significantly larger then the solvent molecules but small enough to show brownian motion [7]. In our case, the particles are dissolved towards concentration significantly lower then the value in which these disolvants are being sold, namely of 2.5% solids (w/v) aqueous suspension [6]. We have used a concentration of 0.1% across all measurements to minimize effects of differing concentration.

To calculate the final viscosity from Eqn. 11 we need to determine the temperature and diffusion coefficient. For determining the temperature we used a room thermometer located withing the room we were doing the experiment in. This thermometer had a precision of $0.1°C$. We calculate the diffusion coefficient by measuring the mean squared displacement $\langle R^2 \rangle$ of particles over time. For this we take a video of our particles using the DinoCapture 2.0 software. We then use the python library trackpy.py to track the movements of the particles recorded in the video. For the tracking to work correctly, the values of several parameters needed to be tailered to each video. An estimate the size of the particles in pixels, called the "feature size", was needed. This was moslty done uniformly across videos of particles with the same diameter. In some cases, however, the focus of the microscope made it so the value had to be tweaked for trackpy to track the particles correctly. We identified this as an artifact of diffraction within our particles where a bright spot would appear in the midle of the particle due to interference. This meant a higher feature size was needed to track the whole particle and not create behaviour where the tracking focused on only a part of the particle. Another important parameter was the so called "minmass". This is a measure of how strong the particle visibility needs to be for it to be tracked. By setting a higher minamss, the particles have to be clearer and in the foreground to be tracked. This helped eliminate accidentally tracking shadows of particles out of focus as well as any dust present in the setup.

To transfer the tracked mean square displacement from pixels to micrometers we needed to measure a calibration factor of how many micrometers was one pixel. For this we used a software called FiJi and a micrometer ruler that we placed under the microcsope with the same magnification as our videos. We then took a picture and measured the distance between stubs on the ruler in pixels, while we knew the physical distance between them. This yielded a calibration factor of $0.130 \pm 0.003 \mu m/px$.

To get the diffusion coefficient from this mean squared dispalcement calculation we used the formula in Eqn. 10. We made a linear fit through the values of mean squared displacement over time to get a slope (derivative), which was $4D$. We then just divided by 4 to get the diffusion

coefficient. To help the linear fit get better data, we incormporated the initial condition that the mean squared dispalcement at time 0 is 0.

The uncertainties that we managed to propagate through this process was the unjcertainty in particle diameter, the uncertainty created by thermometer precision, the uncertainty in the micrometer to pixel conversion and finally, we got an uncertainty for the diffusion coefficient from the linear fit we performed. The uncertainty for the viscosity as well as all previous uncertainty propagations were done using formula in Eqn. 12.

Once the diffusion coefficient was calculated, we used the Stokes-Einstein relation (Eqn. 11) to get the resulting viscosity for a given particle radius and its uncertainty. To average over different viscosities calculated for the same radius we used the Birge ratio method to potentially correct for the mismatch between single-point uncertainties and the spread of viscosity values [8].

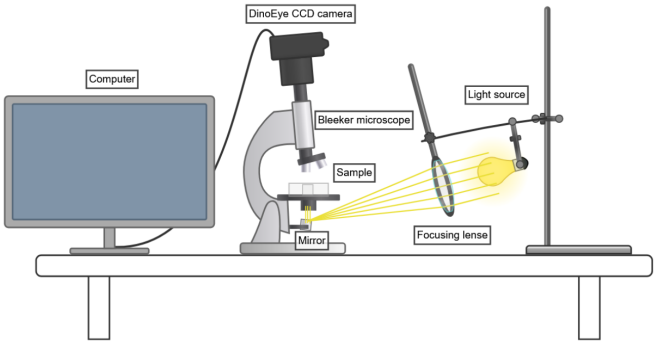


FIG. 1. Schematic showing our experimental setup.

IV. RESULTS

Our meausred value of temperature was of the order of $\sim 21°C$. The diffusion coefficient was of the order $\sim 10^{-1}\mu m$ depending on the particle radius. In the end we calcualted 25 viscosity values, 6 for $r = 0.373$, 6 for $r = 0.53$, 8 for $r = 0.77$ and 5 for $r = 1.03$. We then averaged the results for each radius and the resulting viscosities were $\eta_{r=0.373} = 2.7 \pm 0.1mPa \cdot s, \eta_{r=0.53} = 2.1 \pm 0.2mPa \cdot s, \eta_{r=0.77} = 2.2 \pm 0.2mPa \cdot s, \eta_{r=1.03} = 2.0 \pm 0.4mPa \cdot s$ and a fit of a cosntant function that yields resulting viscosity of $\nu = 2.4 \pm 0.1mPa \cdot s$.

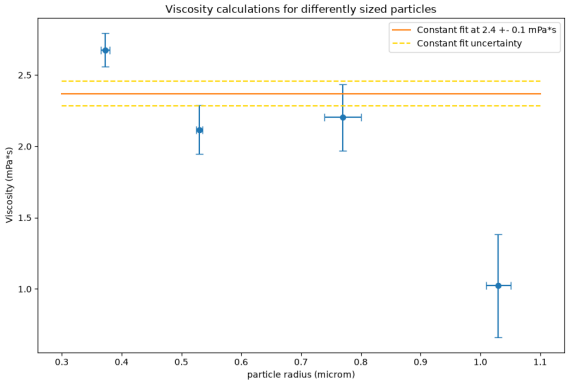


FIG. 2. The final result of average measurement of viscosity for particle radii $r \in \{1.03, 0.77, 0.53, 0.373\}\mu m$ as $\eta_{r=0.373} = 2.7 \pm 0.1mPa \cdot s, \eta_{r=0.53} = 2.1 \pm 0.2mPa \cdot s, \eta_{r=0.77} = 2.2 \pm 0.2mPa \cdot s, \eta_{r=1.03} = 2.0 \pm 0.4mPa \cdot s$ and a fit of a cosntant function that yields resulting viscosity of $\nu = 2.4 \pm 0.1mPa \cdot s$.

V. DISCUSSION

The aquisition and analysis of data for the experiment outlined in this article proved to be rather difficult.

Firstly, the temperature measured by the room thermometer is not necessarily the temperature of our sample. We tried to not touch any components of the smaple more than necessary as to not influence the temperature with our won body heat. That being said, diring the whole measuring procedure we had a light shile direcly on the sample, which could have potentially influenced the temperature within. Unfortunately, we did not have an instrument ot emasure the temperature of our sample directly, but that would be something that could be imporved. One of the problems we encountered was the inconsistency of our setup. Since the stand holding the light and the lense were not fixed in a position relative to the microscope, every time we disassembled and rebuilt the setup, our lighting conditions changed. We tried to reduce this effect as much as possible by placing the microscope and light aparatus into a relative position as close to the initial one as we could, however, this process was not aided by any tools and only eye estimation was used. Unfortunately, we were also not able to do perform this in an efficient way at the beginning of the data aquisition phase, which cost us valubale time. A way to achieve a more consistant setup would be using a microscope with a fixed light or potentially securing the light to the microscope in some way.

Another issue, which we think had a bigger contribution to decreasing the quality of final data, was the tracking of particles using trackpy. Even after trying to mitigate effects disturbing the tracking of the particles, we observed that the effect of wrongful tracking is significant, which we were not able to quantify within the uncertainties. Furthermore, this might be a systematic error.

One of the issues with the tracking of these particles lied in the light interfenece effects that made it so the particles transitioned from dark to light around the focus point. We tried to go around this problem by implementing a slight de-focus for the marticles we wanted to track

to elimiate this problem, but then the particles were naturally less sharp, which introduced a new problem. Even though we did not quantify the difference beween in- and out of focus particles, we believe the de-focused particles produced better data, or were the very least easier to fine-tune the trackpy parameters for.

Another contribution to the difficulty of tracking was the fact that our microscope was set to a specific depth of field, where the particloes we wanted to observe we in the focus we wanted (which as mentioned before was a slight de-focus). However, the sample was not 2D, but the particles also moved in the direction of our line of sight. This meant that sometimes the particles moved further into de-focusing themselves or covnersely, moved into the focus, introducing the interference problem again. In practice this meant that we had to only keep video data that did not contain these issues, which meant taking more videos than needed and filtering those that were deemed unsuable due to these phenomena.

Finally, another obstacle we had to overcome was the fact that the longer we observed a given sample, the higher the calcualted viscosity seemed to climb. We attributed this to the fact that are sample was a liquid, whcih was only trapped between the sample glass plate on either side. This mean, however, that parts of the liquid could evaporate and other processes disturbing the integrity of the sample could take place. To mitigate this phenomen and its effect on our data, we only used a given sample for a limited time after its preperation, wafter which we were forced to make a new one if more measurements were required.

VI. CONCLUSIONS

Conclusion

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